from some point of $\mathcal{D}_1$ to some point of $\mathcal{D}_2$. This condition roughly says that the trajectories starting from some arbitrarily small open set in $\mathcal{D}$ will eventually fill the interior of $\mathcal{D}$ densely. We will see later that this concept, which is essentially equivalent to *ergodicity*, is a crucial component of molecular theories. Below, we will demonstrate, using numerical experiments, that even a simple nonlinear system reminiscent of molecular dynamics can exhibit the hallmarks of a chaotic system. After a short time, small perturbations of the initial conditions will become large, of order of the dimension of the system, and the "memory" of the initial conditions is lost. In this sense the arrangement of atoms rapidly becomes disordered in a similar way to the playing cards of a deck as they are shuffled. The idea of a chaotic dynamical system and its interpretation as a randomizing process is further developed in later chapters. For an informal but valuable introduction to this topic the reader is referred to the short book of Ruelle [318].

*Example 1.9*[7] *(Anisotropic Oscillator)* A perturbation of the central force problem illustrates the consequences of a breakdown of integrability. Consider the system with energy

$$E(x, y, \dot{x}, \dot{y}) = \frac{1}{2}(\dot{x}^2 + \dot{y}^2) + \frac{\kappa(c_3)}{2}(r - l(c_3))^2, \qquad r = \sqrt{x^2 + y^2}, \tag{1.9}$$

where

$$c_3 = \cos(3\theta)$$

is defined in terms of the angular coordinate of the position $(x, y)$ with respect to the $(1, 0)$-direction,

$$\cos\theta = c = \frac{x}{r}, \qquad c_3 = 4c^3 - 3c,$$

and we have defined

$$\kappa(c_3) = \kappa_0(1 - \frac{1}{2}\varepsilon c_3), \qquad l(c_3) = l_0(1 + \frac{1}{2}\varepsilon c_3).$$

Effectively this models the mythical behavior of an atom that binds more tightly to a molecule in certain configurations than in others. In our experiments we took, for simplicity, $l_0 = \kappa_0 = 1$, and considered the effect of varying $\varepsilon$. For $\varepsilon \neq 0$, the angular momentum is not preserved, and the only invariant is the energy.

When $\varepsilon$ is small one finds trajectories have a regular appearance for many values of the initial conditions, typically either a periodic orbit or one which repeatedly and *quasiperiodically* traverses a portion of the domain. As $\varepsilon$ is increased, the orbits

[7] The Anisotropic Oscillator is included as an example in the **MD.M** package, see http://MolecularDynamics.info